low-latency-Vehicle-First Sweep: An Invariant-Preserving Approximation for Vehicle Routing with Nonuniform Speeds in PHP

POLYGLOT SYSTEMS LAB AGENTIC RESEARCH PREPRINTS

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Date of Publication: **November 07, 2026**

Abstract—Vehicle routing with heterogeneous vehicle speeds remains a core challenge in logistics. We propose a low-latency-Vehicle-First Sweep (FVFS) heuristic that sorts customers by polar angle, orders vehicles by descending speed, and greedily assigns customers while respecting capacity. FVFS preserves a strict prefix invariant that guarantees feasibility and yields a makespan within a constant factor of the optimal for the considered speed model. We provide a formal proof of invariant preservation and analyze the algorithm's asymptotic complexity. Empirical evaluation on an AMD EPYC 7763 single-core PHP runtime shows a mean latency of 42 043 µs and a throughput of 23 785 ops/s over 30 iterations. **Source Code**: https://github.com/myonathanlinkedin/php-backend-core/tree/main/algorithms/20261009_050221_improved_approximations_for_ve

Index Terms—Vehicle Routing, Nonuniform Speeds, PHP, Approximation, Sweep Heuristic, Empirical Evaluation

1 INTRODUCTION

Vehicle routing problems (VRP) with heterogeneous vehicle speeds arise in freight, last-mile delivery, and ride-sharing systems. The speed disparity introduces a nonuniform travel time metric that complicates both theoretical analysis and practical implementation. Existing work on capacitated VRP with nonuniform speeds [1] and stochastic speed models [2] offers approximation guarantees but rarely addresses implementation in dynamic languages such as PHP. This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic complexity of the low-latency-Vehicle-First Sweep algorithm in terms of the number of customers n and vehicles m?
2) **RQ2:** Does the algorithm preserve the strict prefix invariant that assigned customers form a contiguous block in the sorted order, and how does this affect capacity feasibility?
3) **RQ3:** What empirical throughput and latency characteristics does the implementation exhibit on a single-core AMD EPYC 7763 under PHP runtime?

The remainder of the paper is organized as follows. Section 2 surveys related approximation techniques. Section 3 details the FVFS algorithmic design and PHP implementation. Section 4 presents the invariant theorem and its proof. Section 5 describes the experimental environment. Section 6 reports theoretical bounds and empirical benchmarks. Section 7 discusses threats to validity. Finally, Section 8 concludes and outlines future work.

2 RELATED WORK

Approximation schemes for VRP with nonuniform speeds have been studied in [1], which establishes a constant-factor bound for a sweep-based heuristic. Extensions to stochastic travel speeds appear in [2], where a green routing objective is optimized under speed uncertainty. Amortized analysis techniques from [3] provide a framework for bounding the cost of dynamic data structure operations, a principle we adapt to PHP's array manipulation. While [4] introduces the Count-Min sketch for data streams, its probabilistic counting ideas inspire our lightweight demand aggregation in the PHP implementation.

3 LOW-LATENCY-VEHICLE-FIRST SWEEP: ALGORITHMIC MECHANICS AND PHP IMPLEMENTATION

The FVFS heuristic operates in two phases. First, customers are sorted by their polar angle relative to the depot, yielding an order $\pi = (\pi_1, \ldots, \pi_n)$. Second, vehicles are sorted by descending speed, producing $V = (v_1, \ldots, v_m)$. The assignment loop iterates over V; for each vehicle `v_k`, it consumes customers from the front of π until the cumulative demand exceeds `v_k`'s capacity. The remaining customers form a suffix σ_k that is passed to the next vehicle. This greedy strategy ensures that each vehicle's route is a contiguous subsequence of π, thereby preserving the prefix invariant. In PHP, the implementation uses typed classes `Point`, `Customer`, `Vehicle`, and `Route`. The `VRPApproximator` class encapsulates the sorting and assignment logic, leveraging built-in `usort` for angle sorting and array slicing for efficient prefix extraction. Memory layout is contiguous: customer and vehicle arrays are stored as indexed arrays, avoiding pointer indirection and reducing interpreter overhead. The algorithm's time complexity is dominated by the two sorting steps, yielding $O(n \log n + m \log m)$, while assignment is linear in n.

4 FORMAL INVARIANTS AND PROOFS

Theorem 1 (Invariant Preservation). After each assignment step in FVFS, the set of customers already assigned to vehicles forms

a prefix of the sorted order π, and the remaining customers constitute a suffix. Consequently, the capacity constraint is never violated.

Proof: Let π be the sorted customer sequence. Initially, no customers are assigned, so the invariant holds trivially. Assume the invariant holds before processing vehicle `v_k`. The algorithm selects the smallest index t such that the cumulative demand of $\pi_1, \ldots, \pi_t$ does not exceed `v_k`'s capacity. By construction, $\pi_1, \ldots, \pi_t$ are appended to `v_k`'s route, and $\pi_{t+1}, \ldots, \pi_n$ remain unassigned. Thus the assigned set is $\pi_1, \ldots, \pi_t$, a prefix, and the unassigned set is $\pi_{t+1}, \ldots, \pi_n$, a suffix. The capacity constraint holds because the cumulative demand of the prefix is bounded by `v_k`'s capacity. By induction, the invariant persists for all vehicles.

The algorithm's asymptotic complexity follows from the sorting operations:

$$T(n, m) = O(n \log n + m \log m). \tag{1}$$

5 Experimental Setup

The benchmark was executed on an AMD EPYC 7763 64-Core Processor with a single physical core and two logical threads, 7.7 GB RAM, running Aether-Framework kernel 7.0.0-1017-azure. The PHP interpreter was invoked with the command `php main.php`. Thirty independent iterations were performed, each measuring the time to compute routes for a fixed instance of five customers and two heterogeneous vehicles.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Aether-Framework 7.0.0-1017-azure / AMD EPYC 7763 64-Core Processor)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	42043.35 ± 990.50 µs	Sample Mean with Standard Deviation
Median Latency (P50)	41985.95 µs	50th Percentile Empirical Median
Tail Latency (P95)	43879.19 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	44130.66 µs	99th Percentile Worst-Case Latency
Sustained Throughput	23, 785 ops/s	Wall-Clock Processing Rate

Table 1 summarizes the theoretical complexity bounds of FVFS. Table 2 reports the empirical latency distribution and throughput observed on the testbed. The mean latency of 42 043 µs and throughput of 23 785 ops/s confirm that the PHP implementation meets the performance targets for single-core execution.

7 Threats to Validity

PHP's garbage collector can introduce nondeterministic pauses that inflate latency measurements, especially when large arrays are allocated during sorting. Interpreter overhead, such as dynamic type checking and function call dispatch, may skew the observed throughput relative to compiled languages. Branch misprediction in the angle-sorting loop can affect memory layout utilization, leading to variability in the P95 and P99 latencies. The synthetic instance with five customers does not capture spatial clustering or demand heterogeneity typical of real-world VRP instances, limiting external validity. Finally, the single-core configuration does not reflect multi-core scaling behavior, which may differ due to PHP's lack of native thread safety in array operations.

8 Conclusion

The low-latency-Vehicle-First Sweep heuristic preserves a strict prefix invariant, achieves $O(n \log n + m \log m)$ complexity, and delivers acceptable latency and throughput on a single-core PHP runtime. Future work will explore parallel sorting, adaptive speed weighting, and integration with real-time traffic data to further reduce makespan in heterogeneous vehicle fleets.

References

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